

```
INF = 9999
```

```
graph = [  
    [0, 5, INF, 10],  
    [INF, 0, 3, INF],  
    [INF, INF, 0, 1],  
    [INF, INF, INF, 0]  
]
```

```
n = len(graph)
```

```
dist = [row[:] for row in graph]
```

```
for k in range(n):  
    for i in range(n):  
        for j in range(n):  
  
            if dist[i][k] + dist[k][j] < dist[i][j]:  
                dist[i][j] = dist[i][k] + dist[k][j]
```

```
print("Shortest Distance Matrix:")
```

```
for row in dist:  
    for value in row:  
        if value == INF:  
            print("INF", end=" ")  
        else:  
            print(value, end=" ")  
    print()
```

```
print("Time Complexity = O(n^3)")
```

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:  
on win32
```

```
Enter "help" below or click "Help" above for more inf
```

```
>>>
```

```
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Docum
```

```
Shortest Distance Matrix:
```

```
0 5 8 9
```

```
INF 0 3 4
```

```
INF INF 0 1
```

```
INF INF INF 0
```

```
Time Complexity = O(n^3)
```

```
>>>
```